Unlock the Lab: evaluation guide

How the workshop works, the six criteria and a glossary

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How the workshop works

‘Chocolate makes you smarter.’ Headlines promise miracles and warn of disasters, but how good is the research behind them?

In this workshop you read short descriptions of studies and judge their quality with six criteria that scientists use: Title, Access, Source, Theory, Methods & Data and Conclusion.

The studies are fictional. They were written for this workshop and are modelled on the strengths and flaws of real research, but none of them describes a real study.

Your two tasks

For each study you give two answers on a scale from 1 (very low quality) to 7 (very high quality):

1. **Predict the crowd.** Guess the average rating that workshop participants give the study. This is the answer that earns points; how it is scored is explained below.
2. **Give your own rating.** Judge the study yourself with the six criteria. Your own rating does not change your score, but it becomes part of the crowd that later participants predict.

After each study you see your score, the target with how it was worked out and the expert’s feedback on each of the six criteria.

How scoring works

Your prediction is scored against a target rating. The target combines the ratings that other participants have given the study so far with an expert’s rating, which counts as if 100 people had given it. Only ratings of the current version of the study descriptions count, and your own rating is never part of your target.

Because the expert counts so heavily, the target stays close to the expert’s rating until many people have taken part. Even with 20 ratings from other participants, the expert’s rating makes up 83% of the target.

A prediction that matches the target exactly scores 100 points. You lose 12 points for each point you are away from the target, so a prediction 1 point away scores 88 and one 2 points away scores 76.

As formulas, where p is your prediction, e the expert’s rating, n the number of other participants who have rated the study and S the sum of their ratings:

$$target = (S + 100 \times e) / (n + 100)$$
$$score = \max(0, 100 - \text{round}(12 \times |p - target|))$$

For example, if the expert rated a study 5 and 20 other participants have given it an average of 3 (a sum of 60), the target is (60 + 100 × 5) / (20 + 100) = 4.67. A prediction of 4 is 0.67 points away, so it scores 100 − round(12 × 0.67) = 92.

Leaderboard and finishing

Your results appear on a public leaderboard under a random nickname, such as ‘Curious Otter 42’. The leaderboard ranks people by their average score per study, once they have rated at least 10 studies.

There are 48 studies. You can stop and see your results at any point after rating 10.

What matters most to you? You have 20 tokens to share between the six criteria you used. Give more tokens to the criteria that matter most to you when you judge a study.

The six criteria

- **Title:** Does the title describe the finding accurately, without hype?
- **Access:** Can anyone read the full study free of charge?
- **Source:** Was it published by a trustworthy, independent outlet?
- **Theory:** Does it build soundly on earlier research and fill a real gap in knowledge?
- **Methods & Data:** Are the design, sample and controls adequate, and are the data open?
- **Conclusion:** Do the claims stay within what the data show?

Quality rubric

Criterion	Low (1–2)	Medium (3–5)	High (6–7)
Title	**Clickbait:** Uses hype such as 'miracle', 'proves', 'cure', 'shocking' or 'breakthrough', or promises far more than the study shows.	**Modest:** Calmly worded, but slightly overstates the finding or leaves out an important limit, such as who was studied or that it shows only a link.	**Accurate:** Boring but true. Says what was studied, in whom and what was found, with any key limit (for example 'Coffee linked to shorter sleep in mice').
Access	**Closed:** Almost no one can read it. The full paper costs money (often £30 or more) with no free copy, or only a summary is public.	**Restricted:** Only some people can read it free. Readers need a university login or a sign-up, or must ask for a copy or know where to find one.	**Open:** Anyone can read the full paper or report free online, with no login or sign-up (open access).
Source	**Suspicious:** Predatory journals (pay to publish, no real peer review), blogs, websites, videos or magazines with no outside checks. Also reports by companies or interest groups on their own products (white papers).	**Use with caution:** Preprints (not yet peer reviewed) and average or less selective journals, or ones keen on eye-catching results. Also peer-reviewed work paid for or checked by people with a stake in the result.	**Trusted:** Selective or well-established peer-reviewed journals (such as those of large scientific societies), independent research bodies or government agencies. No funder has a stake in the result.
Theory	**Unfounded:** Ignores or goes against well-established science, or rests on an idea with no evidence behind it.	**Sound but limited:** Builds reasonably on accepted research, but has a weak point. It rests on an untested assumption, skips other explanations or fills no clear gap.	**Robust:** Uses past research responsibly to fill a clear gap in knowledge. Examples are a new question or a needed replication (repeating a study whose result is still uncertain).
Methods & Data	**Flawed & Closed:** No real study, a tiny sample, hand-picked evidence or a measure that cannot answer the question. No fair comparison (no control group, or groups that differ in obvious ways). The data are kept secret.	**Acceptable:** A reasonable design and sample, but with clear weak points, for example no blinding, a one-off survey or people rating themselves. Data are partly shared or available on request.	**Rigorous & Open:** A strong design and a large enough sample for the question, for example groups chosen by chance, blinding, a preregistered plan and independent checks. Data are open, or shared through a set process that protects privacy.

Criterion	Low (1–2)	Medium (3–5)	High (6–7)
Conclusion	**Overblown:** Claims far more than the evidence shows. It may treat a link as proof of cause or claim effects that were never measured.	**Logical:** Mostly fits the evidence, but plays down limits or leaves out other explanations.	**Nuanced:** Careful. Claims only what the evidence shows and says plainly what the study cannot show.

About the Access criterion

Access is about openness. It asks who can read the full study and whether they must pay, log in or ask for a copy. It does not tell you whether the findings are correct. A free paper can be poor and a paywalled one excellent. So rate Access on openness alone and judge the evidence with the other five criteria.

From six criteria to one rating

Use these six criteria to judge each study, then weigh them up to reach your overall rating from 1 to 7. You can give some criteria more weight than others.

When you rate a study, the app asks: ‘How good is this study overall? Weigh up the six criteria.’

First give each criterion a band: Low, Medium or High. Then start from the band that most criteria share. Mostly Low points to a rating of 1 or 2, mostly Medium to 3 to 5 and mostly High to 6 or 7. The other bands then move you up or down. All Medium is about 4; mostly Medium with one Low is about 3; mostly Medium with one High is about 5. These are rules of thumb, not a formula.

After each study, the expert’s feedback gives a band for each criterion and a summary of how the expert weighed them up, so you can compare your judgement with the expert’s.

At the end you share 20 tokens between the six criteria to show which matter most to you.

A worked example

Take a study of whether a brain-training app improves memory. The title, ‘Brain-training app may boost memory’, leaves out how small the test was (Medium). Readers need a university login to see the full paper (Medium). It appeared in a peer-reviewed journal that accepts most papers it receives (Medium). It builds on memory research but assumes that better scores in the app mean a better memory in daily life (Medium). Only 30 volunteers took part, with no comparison group and no shared data (Low). The authors say the app may help but do not mention that people improve at any test with practice (Medium). That is mostly Medium with one Low, so about 3 is a fair rating.

Glossary

Terms you will meet in the studies.

Actigraphy

Measuring sleep and activity with a motion sensor worn on the wrist. It is more objective than asking people how well they slept.

Anonymised Data

Data from which names and any other details that could identify people have been removed, so that they can be shared while protecting privacy.

Antibody

A protein that the immune system makes to recognise and fight germs. There are several types; IgA, for example, is found in saliva.

Appendix

Extra material at the end of a paper or report, or in a separate online file, such as detailed tables, figures or methods that do not fit in the main text.

Bias

Anything that pushes a study's results away from the truth in one direction, such as how people were chosen, what they or the researchers expected or who paid for the work. Recall bias comes from mistakes in what people remember. In everyday speech, bias also means treating a group unfairly, as in 'gender bias'.

Biotechnology

Using living things, such as bacteria or the genes of plants, to make products. Biotech companies make, for example, medicines or genetically modified (GM) seeds, whose genes have been changed in a lab.

Blinding

Keeping people in a study unaware of who receives which treatment, so that expectations cannot sway the results. In a single-blind study the participants do not know; in a double-blind study neither the participants nor the researchers assessing them know.

Cherry-picking

Presenting only the results or studies that support a claim while leaving out those that do not.

Citation

A reference in one piece of research to another study or source, showing where information came from so that readers can check it. A study is 'cited' each time later research refers to it, and how often a journal's papers are cited is one rough sign of its standing (see Impact Factor).

Code

The computer instructions that researchers write to analyse their data. When the code is shared, others can rerun the analysis and check that they get the same results.

Cohort Study

An observational study that follows a group of people over time, sometimes for decades, to see how their habits or circumstances relate to later outcomes such as illness.

Conflict of Interest

When researchers or their funders have financial, professional or personal interests that could sway the work, such as a company paying for a test of its own product. Researchers should declare these openly, but declaring a conflict does not remove it.

Confounding Variable

A third factor that influences both the supposed cause and the outcome, creating a link between them that is not causal. For example, wealth may explain why people who buy organic food are healthier. Researchers can allow for (adjust for) confounders they have measured, but not for ones they have not.

Control Group

A comparison group that does not get the treatment being tested. Comparing it with the treated group shows whether the treatment itself made a difference.

Correlation vs Causation

Correlation means two things go together, which is often called a link. Causation means one actually causes the other. A correlation alone does not prove causation.

Cross-sectional Study

An observational study that measures people once, at a single point in time. It can show that two things go together but not which came first.

Crossover Trial

A trial in which each participant tries every condition being compared, one after another and usually in random order. Each person then acts as their own comparison.

Data

The information collected during a study, such as measurements, observations or survey answers. Raw data are the individual records; summary data are totals or averages, which others cannot fully recheck.

Data Archive

A public website where researchers store data, code and study plans so that others can download and check them. The Open Science Framework (OSF) is one free example.

Dropout

A participant who leaves a study before the end. If many people drop out, or those who leave differ from those who stay, the results can be biased.

Effect Size

How large a difference or relationship is, not just whether there is one. For the common measure Cohen's *d*, about 0.2 counts as small, 0.5 as medium and 0.8 as large.

Ethics Review

A check by an independent committee that a study protects the people taking part, including their privacy. Researchers may need one before they can use or share sensitive data such as patient records, and it can take months.

Heterogeneity

How much the results of the studies in a review or meta-analysis differ from one another. When it is high, a single pooled figure should be read with caution.

Historical Controls

A comparison group taken from past records, such as earlier patients at the same clinics, instead of people chosen by chance at the same time. The two groups may differ in other ways, such as the care they had or how they were tested.

Hypothesis

A testable prediction about what will happen in a study, based on existing knowledge or theory.

Impact Factor

The average number of times a journal's recent articles are cited by other research. It is a rough sign of a journal's standing but says little about the quality of any single study.

Margin of Error

A range that shows how precise a result is. A result of 40% with a margin of error of 3 points means the true value probably lies between 37% and 43%. Scientists also call this the uncertainty of an estimate.

Matching

Choosing a comparison group that is like the group being studied in key ways, such as age or income. It is used when chance cannot decide who joins which group, but the groups may still differ in ways no one measured.

Meta-analysis

A statistical method that pools the results of several studies on the same question into one overall estimate. It is often part of a systematic review.

Methods

The part of a paper that explains how the study was done: who took part, what was measured and how the data were analysed. The study cards summarise it under ‘Methods & Data’.

Model

A simplified version of something real, often a computer program built from equations and data, used to explain or predict how it behaves. Climate models, for example, estimate future temperatures; statistical models describe patterns in data.

Narrative Review

A review in which the authors choose which studies to discuss, without a search plan set in advance. It can be a useful overview, but it is easier to leave out studies that disagree than in a systematic review.

Observational Study

A study in which researchers record what people already do or experience, without assigning them to groups. It can reveal associations but cannot by itself show cause and effect.

Open Access

Research papers that anyone can read free online, with no paywall, login or subscription. Open access is about who can read a paper, not whether its findings are correct.

Open Data

When researchers make their raw data free for anyone to download, so that others can check the results and run their own analyses.

P-hacking

Trying many different analyses and reporting only those that reach statistical significance. It makes chance findings look real; preregistration helps to prevent it.

Patent

A legal right to an invention, which stops others from making or selling it for a number of years. Researchers may keep details secret until a patent is granted.

Paywall

A barrier that means you must pay to read a paper, often £30 or more. Universities pay subscriptions so that their staff and students can read many papers through a login, but most other people cannot.

Peer Review

Checking of a research paper by other experts in the field before it is published. It catches many problems but not all, and how thorough it is varies between journals.

Placebo

A dummy treatment, such as a sugar pill or a sham device, that looks like the real one but has no active ingredient. Comparing a treatment with a placebo shows whether it works beyond people’s expectations (the placebo effect).

Predatory Journal

A journal that charges authors a fee but skips real peer review. Many trustworthy open-access journals also charge fees; the warning sign is the missing quality control, such as very fast publication or no named editors. These journals put profit before scientific quality.

Preprint

An early version of a research paper shared publicly before peer review, often on a preprint server (a public website for preprints). It lets research spread quickly, but experts have not yet checked it.

Preregistration

Publicly recording a study's plan, including what will be measured and how the data will be analysed, before the data are collected or analysed. Readers can then see whether researchers changed their questions or analyses after seeing the results.

Press Release

A short statement written by a university, journal or company to get news coverage for a study. It is not the study itself and may leave out its limits.

Randomised Controlled Trial (RCT)

An experiment in which chance decides whether each participant receives the treatment being tested or joins a control group. Because the groups start out alike, differences in outcome can be attributed to the treatment.

Regulator

A government body that checks that products such as medicines and vaccines are safe, before and after they go on sale.

Replication

Repeating a study to see if the same results occur. Results that hold up when repeated are more trustworthy than one-off results, and a replication is most needed when an earlier result is new or uncertain.

Results

What a study found, such as measurements and the outcomes of statistical tests, before the authors explain what they think it means. On the study cards, the main results usually appear under 'Conclusion', next to the authors' claims.

Reverse Causality

When the supposed effect is actually the cause. For example, people who sleep badly may use their phones more because they are awake, rather than phone use causing poor sleep.

Sample Size

The number of participants (*N*) or observations in a study. Larger samples generally provide more reliable results.

Single-arm Trial

A trial in which every participant gets the treatment and there is no comparison group. It is often used for early tests of new treatments, but it cannot show how patients would have done without the treatment.

Statistical Significance

A finding is called statistically significant when results at least this strong would rarely occur by chance if there were really no effect (by convention, $p < 0.05$). It does not show how large or important an effect is, and a non-significant result does not prove that there is no effect.

Systematic Review

A review that searches for all the studies on a question using a method set out in advance, assesses their quality and summarises the evidence. Unlike a narrative review, it is designed to avoid cherry-picking.

Theory

In science, a theory is a well-tested explanation of how something works, supported by evidence from many studies; in everyday speech the word often just means a guess. The rubric's Theory criterion asks something different: how soundly a study builds on earlier research and whether it fills a real gap in knowledge, for example with a needed replication.

Variable

Any factor that can change or be measured in a study, such as age, temperature or test scores. A proxy variable stands in for another one: a postcode, for example, can hint at someone's income.

White Paper

A report published directly by a company, industry body or campaign group, usually without independent peer review. White papers are often used for marketing or lobbying.

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